

	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering An Autonomous Institute (Permanently affiliated to Savitribai Phule Pune University)	SEMESTER- VI
ASSIGNMENT- 3		
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1. Objective

The objective of this assignment is to design and develop a dynamic e-commerce web application using HTML, CSS, and JavaScript. The aim is to create an interactive shopping interface where users can view products, add items to the cart, and dynamically interact with the webpage. Bootstrap CDN is used to enhance responsiveness and user interface design.

2. Learning Outcomes

After completing this assignment, I was able to:

- Build a dynamic web application using JavaScript.
- Implement DOM manipulation and event handling.
- Design a responsive UI using Bootstrap.
- Create features like Add to Cart and filtering.
- Improve UI/UX design skills for real-world applications.

3. Brief Theory

An e-commerce web application allows users to browse products and interact dynamically with the interface.

HTML provides structure, CSS enhances styling, and JavaScript makes the page dynamic by updating content without reloading.

Key concepts used:

- **DOM Manipulation** – dynamically displaying products
- **Event Handling** – button clicks (Add to Cart)
- **Arrays & Objects** – storing product data
- **Bootstrap** – responsive grid and styling
- **UI/UX Principles** – clean layout, hover effects, spacing

4. Program Code and Output

Program Code:

```
<!doctype html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <title>Fashion Store</title>


    <!-- Bootstrap CDN -->

    <link

      href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css

      " rel="stylesheet"

    />

    <!-- Font Awesome for icons -->

    <link

      rel="stylesheet"

      href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css

      " />
```

<style>

body {

background-color: #f8f9fa;

font-family: "Segoe UI", Tahoma, Geneva, Verdana, sans-serif;

}

.navbar {

box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);

}

.navbar-brand {

font-weight: bold;

font-size: 1.5rem;

}

.hero-section {

background: url("https://images.unsplash.com/photo-1483985988355-763728e1935b?ixlib=rb4.0.3&ixid=M3wxMjA3fDB8MHxwaG90by1wYWdlfHx8fGVufDB8fHx8fA%3D%3D&auto=for mat&fit=crop&w=1170&q=80")

no-repeat center center;

background-size: cover;

color: white;

padding: 100px 0;

text-align: center;

}

```
.hero-section h1 {  
  font-size: 3.5rem;  
  font-weight: bold;  
  
  text-shadow: 2px 2px 4px rgba(0, 0, 0,  
0.5); }
```

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```
.card {  
  transition:  
    transform 0.3s ease,  
    box-shadow 0.3s ease;  
  
  border: none;  
  
  border-radius: 10px;  
  
  overflow: hidden;  
}
```

```
.card:hover {  
  transform: translateY(-10px);  
  
  box-shadow: 0 10px 20px rgba(0, 0, 0,  
0.15); }
```

```
.card-img-top {  
  height: 300px;  
  
  object-fit: cover;  
}
```

```
.card-body {  
  
  background-color: #ffffff;  
  
}
```

```
.card-title {  
  
  font-weight: 600;  
  
}
```

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```
.card-text {  
  
  color: #0d6efd;  
  
  font-weight: bold;  
  
  font-size: 1.2rem;  
  
}
```

```
.cart-badge {  
  
  background-color:  
  
  #dc3545; color: white;  
  
  padding: 0.25em  
  
  0.6em; border-radius:  
  
  50%;  
  
  font-size: 0.8rem;  
  
  vertical-align: top;  
  
}
```

```
.btn-primary {
```

```
background-color:
#0d6efd; border: none;
}
```

```
.btn-primary:hover {
background-color:
#0b5ed7; }
```

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```
footer {
background: #212529;
color: white;
text-align: center;
padding: 20px;
margin-top: 40px;
}
</style>
</head>
<body>
<!-- Navbar -->
<nav class="navbar navbar-expand-lg navbar-light bg-light
sticky-top"> <div class="container">
<a class="navbar-brand" href="#">Fashion Store</a>
<div class="d-flex align-items-center">
<input
id="search-bar"
```

```

        class="form-control me-2"

        type="search"

        placeholder="Search products..."

        aria-label="Search"

    />

    <button

        class="btn btn-outline-dark"

        type="button"

        data-bs-toggle="modal"

        data-bs-target="#cartModal"

    >

        <i class="fas fa-shopping-cart"></i>

        <span id="cart-count"

            class="cart-badge">0</span> </button>

    </div>

</div>

</nav>

<!-- Hero Section -->

<div class="hero-section">

    <div class="container">

        <h1>New Season Arrivals</h1>

        <p class="lead">Check out all the new trends for

            2026</p> </div>

    </div>

```

<!-- Filter -->

<div class="container mt-4">

<div class="row justify-content-center">

<div class="col-md-6">

<select id="filter" class="form-select">

<option value="all">All Categories</option>

<option value="men">Men's Fashion</option>

<option value="women">Women's

Fashion</option> <option value="kids">Kids'

Fashion</option>

</select>

</div>

</div>

</div>

<!-- Products -->

<div class="container mt-4">

<div class="row" id="product-list"></div>

</div>

<!-- Cart Modal -->

<div

class="modal fade"

id="cartModal"

tabindex="-1"

aria-labelledby="cartModalLabel"

aria-hidden="true"

>

<div *class*="modal-dialog modal-lg">

<div *class*="modal-content">

<div *class*="modal-header">

<h5 *class*="modal-title" *id*="cartModalLabel">Your Shopping

Cart</h5> <button

type="button"

class="btn-close"

data-bs-dismiss="modal"

aria-label="Close"

></button>

</div>

<div *class*="modal-body" *id*="cart-items">

<!-- Cart items will be injected here -->

</div>

<div *class*="modal-footer">

<h5 *class*="me-auto">Total: <span

id="cart-total">₹0</h5> <button

type="button"

class="btn btn-secondary"

data-bs-dismiss="modal"

>

Close

</button>

```
<button type="button" class="btn btn-primary"
  onclick="checkout()"> Checkout
```

```
</button>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<!-- Footer -->
```

```
<footer>
```

```
<p>© 2026 Fashion Store. All Rights Reserved.</p>
```

```
</footer>
```

```
<!-- Bootstrap JS Bundle (includes Popper) -->
```

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```
<script
```

```
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
```

```
<script>
```

```
const products = [
```

```
{
```

```
  id: 1,
```

```
  name: "Men's Classic T-Shirt",
```

```
  price: 499,
```

```
  category: "men",
```

```
  img:
```

```
  "https://images.unsplash.com/photo-1521572163474-6864f9cf17ab?w=500", },
```

```
{  
  id: 2,  
  name: "Women's Summer Dress",  
  price: 899,  
  category: "women",  
  img:  
    "https://images.unsplash.com/photo-1572804013427-4d7ca7268217?w=500", },
```

```
{  
  id: 3,  
  name: "Men's Denim Jeans",  
  price: 1199,  
  category: "men",  
  img:  
    "https://images.unsplash.com/photo-1602293589930-45a9ecba3ab5?w=500", },
```

```
{  
  id: 4,  
  name: "Women's Knit Top",  
  price: 699,  
  category: "women",  
  img:  
    "https://images.unsplash.com/photo-1554284126-aa88f22d8b74?w=500", },
```

```
{  
  id: 5,  
  name: "Kid's Graphic Tee",  
  price: 399,  
  category: "kids",
```

```
img:
"https://images.unsplash.com/photo-1619215993198-688598642c6a?w=500", },
{
id: 6,
name: "Men's Casual Shirt",
price: 999,
category: "men",
img:
"https://images.unsplash.com/photo-1618354691373-d851c5c3a990?w=500", },
{
id: 7,
name: "Women's Floral Skirt",
price: 799,
category: "women",
img:
"https://images.unsplash.com/photo-1589465885857-add714393a7c?w=500", },
{
id: 8,
name: "Kid's Play Trousers",
price: 599,
category: "kids",
img:
"https://images.unsplash.com/photo-1595333232630-c46e48595146?w=500", },
];
```

```
let cart = [];
```

```
function displayProducts(filter = "all", searchTerm = "") {
```

```
  const list = document.getElementById("product-list");
```

```
  list.innerHTML = "";
```

```
  const filtered = products
```

```
    .filter((p) => filter === "all" || p.category === filter)
```

```
    .filter((p) =>
```

```
      p.name.toLowerCase().includes(searchTerm.toLowerCase()),
```

```
    );
```

```
  if (filtered.length === 0) {
```

```
    list.innerHTML = `
```

```
    found.</p>`; return;
```

```
  }
```

```
  filtered.forEach((product) => {
```

```
    list.innerHTML += `
```

```
      <div class="col-lg-3 col-md-4 col-sm-6 mb-4">
```

```
        <div class="card h-100">
```

```
           <div class="card-body d-flex flex-column">
```

```
            <h5 class="card-title">{product.name}</h5>
```

```
            <p class="card-text mt-auto">₹{product.price}</p>
```

```

        <button class="btn btn-primary mt-2"
            onclick="addToCart(${product.id})"> <i class="fas fa-cart-plus"></i>
            Add to Cart
        </button>
    </div>
</div>
</div>
`
});
}

```

```

function addToCart(id) {
    const product = products.find((p) => p.id === id);
    const cartItem = cart.find((item) => item.id === id);

    if (cartItem) {
        cartItem.quantity++;
    } else {
        cart.push({ ...product, quantity: 1 });
    }

    updateCart();
}

```

```

function updateCart() {
    const cartCount = document.getElementById("cart-count");
    const cartItemsContainer = document.getElementById("cart-items");
}

```

```
const cartTotal = document.getElementById("cart-total");
```

```
cartCount.innerText = cart.reduce(  
  (sum, item) => sum + item.quantity,  
  0,  
);
```

```
if (cart.length === 0) {  
  cartItemsContainer.innerHTML = `empty.</p>`; cartTotal.innerText = "₹0";  
  return;  
}
```

```
cartItemsContainer.innerHTML = "";
```

```
let total = 0;
```

```
cart.forEach((item) => {
```

```
  total += item.price * item.quantity;
```

```
  cartItemsContainer.innerHTML += `
```

```
    <div class="d-flex justify-content-between align-items-center mb-3">
```

```
      <div class="d-flex align-items-center">
```

```
        
```

```
      <div>
```

```
        <h6 class="mb-0">${item.name}</h6>
```

```
        <small class="text-muted">₹${item.price} x ${item.quantity}</small>
```

```
      </div>
```

```

    </div>

    <button class="btn btn-outline-danger btn-sm"

        onclick="removeFromCart(${item.id})"> <i class="fas fa-trash"></i>

    </button>

</div>

`;

});

cartTotal.innerHTML = `₹${total}`;

}

function removeFromCart(id) {

    cart = cart.filter((item) => item.id !== id);

    updateCart();

}

function checkout() {

    if (cart.length === 0) {

        alert("Your cart is empty. Please add items before checking out.");

        return;

    }

    alert("Thank you for your order! Your items will be shipped soon.");

    cart = [];

    updateCart();

    // Close the modal after checkout

    const cartModal = bootstrap.Modal.getInstance(

        document.getElementById("cartModal"),

```



```
);  
  
cartModal.hide();  
  
}
```

```
document.getElementById("filter").addEventListener("change", (e) =>  
  
  { displayProducts(  
  
    e.target.value,  
  
    document.getElementById("search-bar").value,  
  
  );  
  
});
```

```
document.getElementById("search-bar").addEventListener("input", (e) =>  
  
  { displayProducts(  
  
    document.getElementById("filter").value,  
  
    e.target.value,  
  
  );  
  
});
```

```
// Initial Load
```

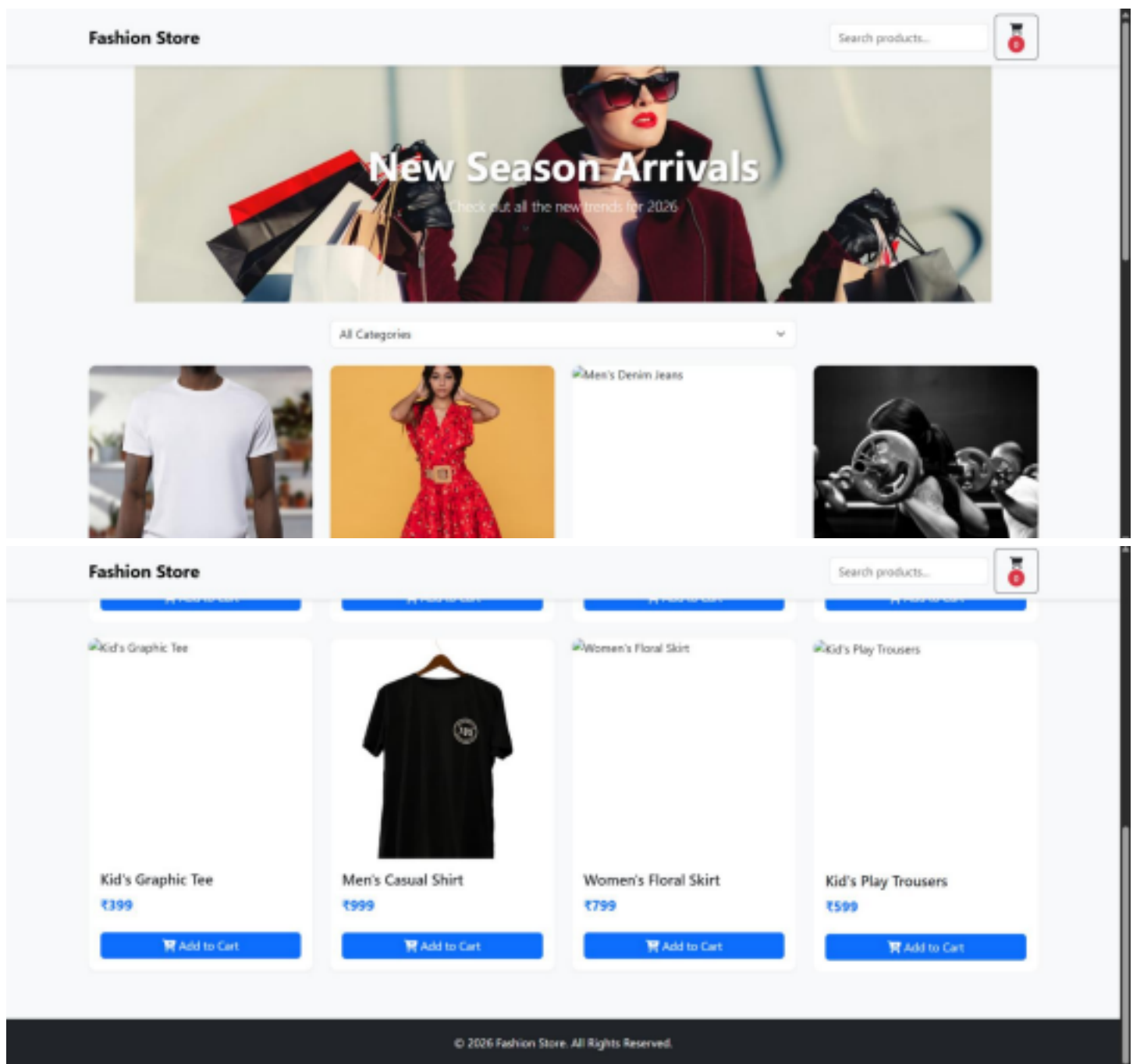
```
displayProducts();
```

```
</script>
```

```
</body>
```

```
</html>
```

Output Screenshot:



5. Conclusion

In this assignment, I developed a dynamic e-commerce web application using HTML, CSS, JavaScript, and Bootstrap. I implemented features like product display, category filtering, and add-to cart functionality using DOM manipulation and event handling. This assignment enhanced my understanding of dynamic web development and improved my UI/UX design skills.